Lopcombe Corner — Peak Level vs Winter Rainfall (Oct–Mar, corrected)

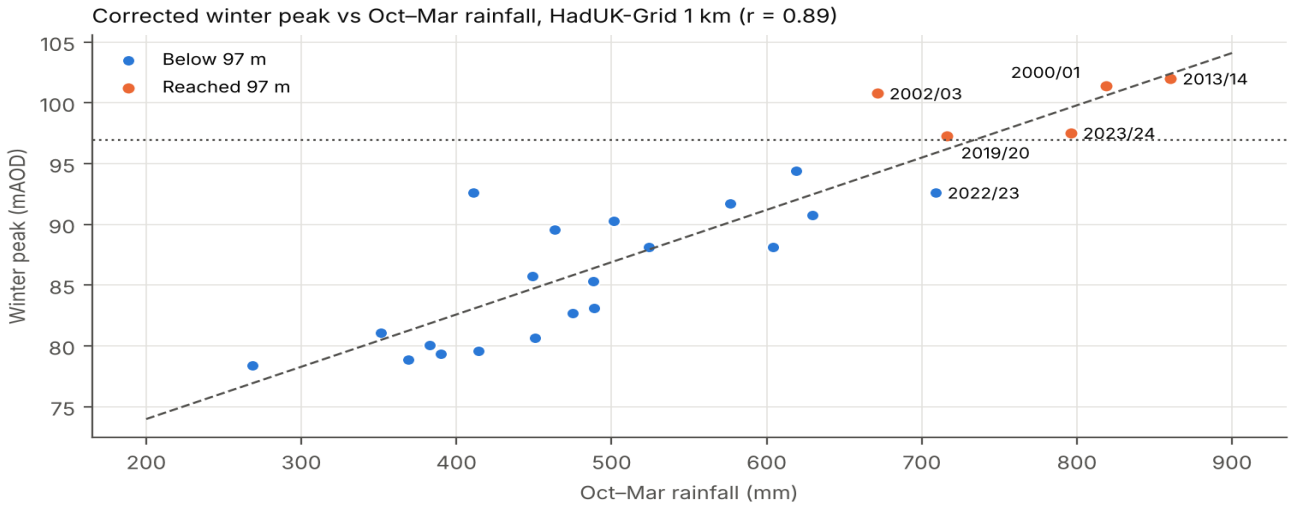
25 winters 2000/01–2024/25 · corrected truncation-proof winter peaks · rainfall re-summed by month from Broughton E13580 and validated against Tim Vaughan's Broughton figures · analysis 7 Sep 2026

Recreated 20 Oct 2026. Statistics below are those from the September analysis. The monthly Broughton table embedded in the original PDF was not kept in the project records, so the data table here pairs the corrected peaks with **HadUK-Grid 1 km rainfall** for the borehole's grid cell (extracted 20 Oct 2026), which supersedes the Broughton sums. With HadUK-Grid, peak vs Oct–Mar is r = 0.89 for these 25 winters and r = 0.82 over 61 winters back to 1964.

Headline — Oct–Mar is the right window; antecedent low matters too

Window	r	r^2	+ antecedent low
Oct–Dec	0.72	0.52	—
Oct–Jan	0.80	0.64	R^2 0.73
Oct–Mar	**0.86**	**0.74**	**R^2 0.83**

- Extending the window from Oct to Mar lifts r from 0.80 to 0.86: the recharge that sets the peak runs into Feb–Mar.
- Adding the antecedent recession low lifts Oct–Mar R^2 to 0.83 — about +5 m of peak per +1 m of antecedent level — so the buffer effect is a clear contributor once the rainfall window is right.
- Oct–Mar fit (Broughton): peak ≈ 0.047 × rain + 64.4, about +4.7 m per +100 mm.
- Why Oct–Jan fell short: most peaks occur Feb–Mar, fed by late-winter rain — Feb 2020 (148 mm) drove the 24 Feb 2020 peak of 97.25 m; Feb 2014 (141 mm) the 13 Feb record of 102.0 m.



97 m Pitton threshold (revised)

Five of 25 logger-era winters reached 97 m (about 1 in 5): 2000/01, 2002/03, 2013/14, 2019/20 and 2023/24. On the Broughton sums all had Oct–Mar ≥ ~615 mm and none below ~565 mm reached it; on HadUK-Grid all had ≥ ~670 mm. An Oct–Mar running total is a better early indicator than Oct–Jan, but timing and antecedent state still shift the exact peak — 2022/23 was very wet yet peaked at only 92.6 m.

Validation and caveats

- Broughton monthly Oct–Jan sums vs Tim Vaughan: 2001 529 vs 534, 2003 555 vs 557 (~1%), but 2014 565 vs 605, 2020 445 vs 484, 2024 456 vs 509 (7–11% low) — the fetch under-added the wettest months. HadUK-Grid gives 578, 603, 634, 496 and 505 mm.
- Winter 2018/19 Broughton totals under-read (12-day data gap in January 2019).
- Broughton is about 10 km from the borehole; HadUK-Grid uses the borehole's own 1 km cell.

Data: corrected winter peaks and HadUK-Grid 1 km rainfall

Winter	Oct–Dec mm	Oct–Jan mm	Oct–Mar mm	Peak mAOD	Peak date
2000/01	**480**	**578**	**819**	**101.39**	
2001/02	229	328	488	85.33	
2002/03	**511**	**604**	**671**	**100.79**	
2003/04	278	394	475	82.72	
2004/05	263	312	382	80.05	
2005/06	245	272	390	79.33	
2006/07	373	466	630	90.76	
2007/08	194	311	449	85.76	
2008/09	198	292	411	92.62	
2009/10	387	457	604	88.15	
2010/11	170	272	351	81.05	
2011/12	174	222	268	78.40	approx.
2012/13	398	496	619	94.42	
2013/14	**402**	**634**	**860**	**102.01**	**13 Feb 2014**
2014/15	312	412	489	83.13	
2015/16	221	359	524	88.15	
2016/17	159	247	369	78.86	
2017/18	210	295	464	89.54	
2018/19	268	305	451	80.66	
2019/20	**380**	**496**	**716**	**97.25**	**24 Feb 2020**
2020/21	384	474	576	91.72	13 Feb 2021
2021/22	247	272	414	79.61	
2022/23	429	559	709	92.62	21 Jan 2023
2023/24	**420**	**505**	**796**	**97.54**	**12 Jan 2024**
2024/25	270	415	501	90.31	9 Dec 2024

Bold rows reached the 97 m threshold. Rainfall: Met Office HadUK-Grid v1.3.2.ceda (release v20260512), cell centred E 425500 / N 135500. 2011/12 peak approximate (drought year).